

Original Article

The effect of pre-treatment with transcutaneous electrical acupoint stimulation on the quality of recovery after ambulatory breast surgery: a prospective, randomised controlled trial

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Summary

Electroacupuncture has been demonstrated to be effective at alleviating pain and postoperative side-effects. Our aim was to investigate whether transcutaneous electric acupoint stimulation, a low-skill alternative to needle-based electroacupuncture, could improve the quality of recovery after ambulatory surgery. Seventy-two women scheduled for cosmetic breast surgery were randomly allocated to transcutaneous electric acupoint stimulation or sham groups. Patients in the transcutaneous electric acupoint stimulation group received 30 min of electrical stimulation at three acupoints located on the hand and forearm before the induction of general anaesthesia. We found significant mean (SD) differences between the transcutaneous electric acupoint stimulation and sham groups in the mean (SD) length of recovery room stay (35.6 (12.9) min vs 48.3 (16.3) min, $p = 0.01$), time to removal of the laryngeal mask airway (10.2 (2.5) min vs 17.8 (4.4) min, $p = 0.01$), and time to reorientation of the patient (14.6 (3.2) min vs 26.5 (5.0) min, $p = 0.01$). Further, postoperative pain scores and the incidence of side-effects were all lower in the transcutaneous electric acupoint stimulation group. In conclusion, transcutaneous electric acupoint stimulation can significantly improve the quality of recovery and decrease the incidence of anaesthesia-related side-effects for patients undergoing ambulatory surgery.

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Introduction

Ambulatory surgery is increasingly popular in many countries. Currently, 60–70% of the surgical procedures in the USA each year are performed in an ambulatory setting [1, 2]. Patients need less postoperative care and benefit from lower complication rates, a better cost-benefit ratio, and earlier resumption of professional and social activities [3]. The length of recovery room stay and the control of postoperative anaesthesia-related

side-effects are crucial to ensure patient safety and improve the quality of recovery [4]. Therefore, to shorten recovery room stay, improve postoperative pain relief and reduce postoperative side-effects, an integrated approach combining pharmacological methods and various complementary and alternative analgesic techniques has been recommended [5].

Acupuncture is a traditional Chinese medical technique that involves the insertion of needles at acupoints

to treat diseases by Jingluo (the system of meridians, through which energy is thought to flow through the body in Chinese medicine). It is usually applied to relieve pain. Many operations performed under acupuncture anaesthesia or combined with narcotic drugs have been reported since the 1970s [6], but there has been increasing interest in the use of acupuncture or acupuncture-related techniques in combination with general anaesthesia; the focus has shifted towards using acupuncture to reduce intra-operative anaesthetic and opioid requirements [7]. Several clinical trials have been conducted to evaluate the effect of electroacupuncture on the consumption of intra-operative anaesthetics and on drug-related side-effects, with promising results [8–10]. Moreover, many studies conducted in the USA and Europe suggest that acupuncture appears to be effective for postoperative nausea and vomiting [11, 12]. However, the effects of acupuncture or electroacupuncture on recovery room stay and the quality of recovery from anaesthesia remain unknown.

Compared with acupuncture or electroacupuncture, transcutaneous electric acupoint stimulation (TEAS) is a non-invasive technique and has some advantages, including no risk of infections or needle-induced contagious disease and reduced fear of stimulation. It can potentially be applied by any anaesthetist or pre-operative personnel with minimal training. Thus, we conducted this controlled, double-blinded clinical trial to investigate whether pre-operative TEAS could shorten recovery room stay or reduce the incidence of anaesthesia-related side-effects.

Methods

The trial was approved by the Xijing Hospital Research Ethics Committee and is registered as NCT01700855 at clinicaltrials.gov. All participating patients gave signed informed consent before being enrolled in this study.

We conducted this prospective, randomised, placebo-controlled clinical trial in Xijing Hospital at the Fourth Military Medical University. Seventy-two women aged 20–50 years, of ASA physical status 1 or 2, scheduled for elective cosmetic breast surgery, were eligible for this trial, which took place between July and December 2012. The exclusion criteria were as follows: heart, lung, liver, kidney or endocrine system

disease; mental disorder, sore throat; obesity; potentially difficult airway; or previous experience with acupuncture treatment.

Patients were assigned to either the TEAS group or the sham group using a randomisation sequence based on a table of randomly generated numbers. The patients were not informed of the allocation, and blinding was assured for other persons involved in the study by using placebo gel electrodes applied to the same acupressure points in patients in the sham group, as described previously [13]. To enhance the reliability of data acquisition, both the randomisation and allocation lists were concealed from the anaesthetists who gave the general anaesthetic to study patients, and from the persons who performed the final statistical analysis.

Patients in the TEAS group received pre-operative TEAS by an experienced acupuncturist for 30 min before the induction of anaesthesia in the operating theatre. Transcutaneous electric acupoint stimulation was applied to three pairs of acupoints: bilateral Hegu (LI4); Neiguan (PC6); and Zusanli (ST36). These acupoints were identified according to the traditional anatomical localisations (Fig. 1). The acupuncturist carried out this stimulation using a dense-disperse frequency of 2/10 Hz with an intensity of 6–9 mA for 30 min using the Hwato electronic acupuncture treatment instrument (model No. SDZ-V; Suzhou Medical Appliances Co., Ltd., Suzhou, China). The optimal intensity was adjusted to maintain a slight twitching of the regional muscle according to individual maximum tolerance. In the sham group, the patients were connected to the instrument, but electronic stimulation was not applied.

On arrival in the operative room, standard anaesthetic monitors, including electrocardiogram, non-invasive arterial blood pressure, pulse oximetry, end-tidal carbon dioxide partial pressure ($P_{\text{E}}\text{CO}_2$), bispectral index (BIS) monitoring and body temperature were applied. Patients in both groups received midazolam 0.03 mg.kg^{-1} as soon as intravenous access was established. Anaesthesia was induced intravenously with fentanyl $3.0 \text{ }\mu\text{g.kg}^{-1}$ and propofol 2.0 mg.kg^{-1} . Rocuronium 0.5 mg.kg^{-1} was given to facilitate laryngeal mask airway (LMA) insertion. After insertion, mechanical ventilation was used throughout surgery with a tidal volume of 8 ml.kg^{-1} to maintain $P_{\text{E}}\text{CO}_2$

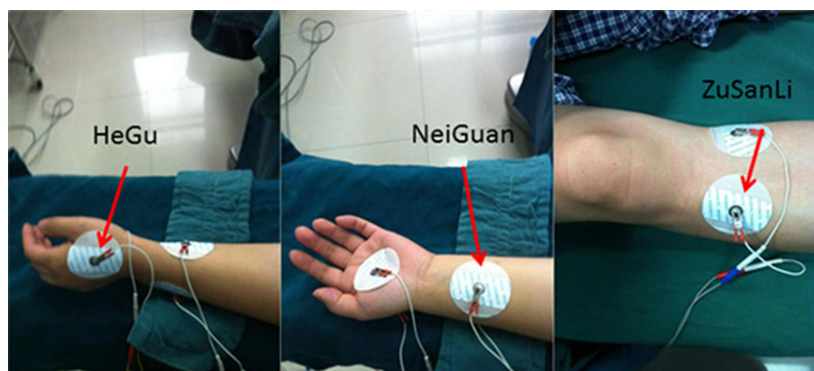


Figure 1 Location of Hegu (LI4), Neiguan (PC6) and Zusanli (ST36) acupoints.

at 4.5–6.0 kPa. Anaesthesia was maintained with target-controlled infusions of propofol [14] and remifentanyl [15]. The anaesthesiologist adjusted the effect-site concentration of propofol and remifentanyl according to the haemodynamics and bispectral index. All the operations were performed by the same surgeon according to a standard protocol. The surgeon did not inject vasoconstrictors or local anaesthetic drugs into the operative site.

Heart rate, mean arterial pressure, electrocardiogram, $P_{\text{E}}\text{CO}_2$ and pulse oximetry were recorded in both groups before the onset of TEAS (baseline, T_0), at the end of TEAS (T_1), at the beginning of surgery (T_2), 30 min after the start of surgery (T_3), at the end of surgery (T_4) and 5 min after removal of the LMA (T_5). All anaesthetic infusions were stopped 5 min before the end of surgery. Meanwhile, parecoxib sodium (40 mg) and tropisetron hydrochloride (4 mg) were given for prophylaxis of postoperative pain and nausea/vomiting, respectively. The patients were transferred to the recovery room after removal of the LMA.

The primary outcome of this study was the length of recovery room stay. The secondary endpoints were the consumption of anaesthetics, the time to removal of the LMA, the time to reorientation and postoperative side-effects (incidence of respiratory depression, nausea, vomiting, dizziness and pruritus).

After recovery room admission, the nurses manually recorded the patient's respiratory rate, $S_{\text{p}}\text{O}_2$, mean arterial pressure, heart rate, levels of pain and sedation, as well as dizziness, pruritus, nausea and vomiting at 10-min intervals. Pain was assessed using an 11-point verbal analogue scale where 0 = no pain; 1–3 = mild pain; 4–6 = moderate pain; 7–9 = severe pain and

10 = intolerable pain. The level of consciousness was evaluated by a sedation scale [16], in which 1 = fully awake; 2 = occasionally sleepy; 3 = asleep but responds to verbal command; 4 = asleep but responds to mild physical stimulation and 5 = asleep and difficult to rouse. Pain and sedation scores were also recorded 4, 8 and 24 h postoperatively.

Time to reorientation was defined as recollection of the current date, the patient's date of birth and their surgeon's name. Respiratory depression was defined as respiratory rate < 8 breaths.min⁻¹. Nausea was defined as a subjectively unpleasant sensation associated with awareness of the urge to vomit, and vomiting was defined as the forceful expulsion of gastric contents from the mouth brought about by powerful sustained contraction of the abdominal muscles. Dizziness was defined as a sensation of spinning or having one's surroundings spin about one.

Discharge readiness was assessed by the recovery room nurse using the modified Aldrete score [17] every 10 min. Patients were considered ready for discharge when the modified Aldrete score was 10 and the pain score was < 5 . Recovery room nursing personnel were blinded to patient study groups and intra-operative treatment.

Concentrations of adrenaline, noradrenaline, adrenocorticotrophic hormone (ACTH), glucose and cortisol were measured as markers of stress levels, using commercially available immunosorbent kits. Blood glucose was measured by glucose oxidase. Peripheral venous blood samples were collected into pre-cooled anticoagulant tubes containing trasylol at four time points, T_1 to T_4 as previously defined. The samples were centrifuged at $3000 \times g$ for 10 min. Serum samples

were separated into polyethylene tubes and stored at -80°C until analysis.

For our primary outcome measure, the mean (SD) time to achieve eligibility for discharge from the recovery room was 50 (20) min in a pilot study. To capture an anticipated 30% decrease in the primary outcome measure to 35 (15) min, a sample of 46 patients (23 per group) would provide a two-sided unpaired Student's *t*-test with 80% power at an alpha of 0.05. To account for potential loss to follow-up and enable greater statistical power for secondary analyses, the sample size was increased to 72 patients (36 per group).

All statistical analyses were performed using SPSS software version 18.0 (SPSS Inc., Chicago, IL, USA). All continuous variables were compared using unpaired Student's *t*-tests or chi-squared tests as

appropriate. The Kruskal–Wallis test with post-hoc multiple comparisons was used to compare the values between the groups at each time point. The level of significance for all statistical tests was set at 0.05.

Results

As shown in Fig. 2, a total of 72 patients were enrolled in the study; seven patients (9.7%) dropped out of this study because they declined to undergo TEAS (3) or met the exclusion criteria (4). Sixty-five patients' records were analysed.

The patients' characteristics and type and durations of procedures were similar between the two groups (Table 1).

There were significant differences between the two groups in total dose of remifentanyl given, the time to removal of the LMA, the time to reorientation and the

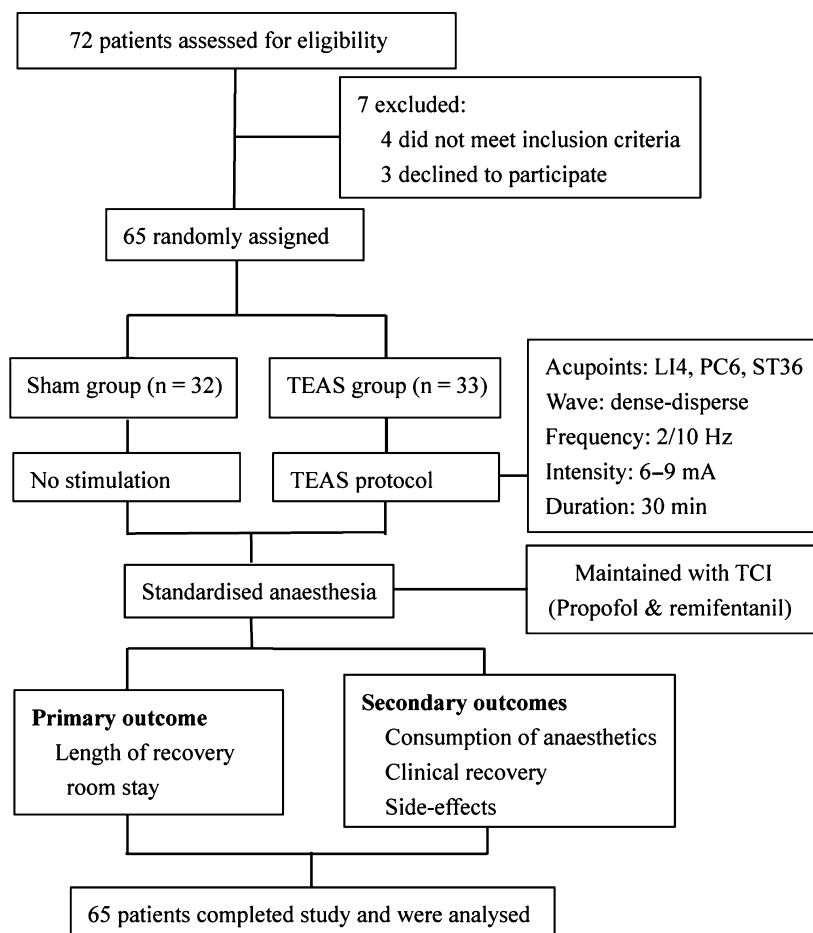


Figure 2 Trial flowchart showing the study procedures and the number of patients. TEAS, transcutaneous electrical acupoint stimulation; TCI, target-controlled infusion.

Table 1 Characteristics and surgical details of patients receiving transcutaneous electrical acupoint stimulation (TEAS) or sham treatment. Values are mean (SD) or number (proportion).

	TEAS (n = 33)	Sham (n = 32)
Age; years	35.0 (7.6)	34.3 (9.1)
BMI; kg.m ⁻²	22.5 (1.2)	22.2 (1.1)
Operation		
Breast augmentation	11 (33.3%)	9 (28.1%)
Reduction mammoplasty	12 (36.4%)	10 (31.3%)
Ogilvy pull out	7 (21.1%)	9 (28.1%)
Other	3 (9.1%)	4 (12.5%)
Duration of operation; min	89 (30)	90 (31)
Duration of anaesthesia; min	115 (32)	120 (30)

length of recovery room stay, all favouring the TEAS group (Table 2). All patients were discharged from recovery room according to the criteria prescribed by the protocol. There were three patients in the TEAS group who fulfilled the criteria for discharge within 15 min, compared with only one in the sham group. Overall, 12 patients' recovery room stay exceeded 1 h, including three in the TEAS group and nine in the sham group.

Pain scores were significantly lower at each time point until 24 h postoperatively in the TEAS group. Sedation scores were significantly different only at 4 h after operation (Fig. 3). Dizziness, pruritus, nausea and vomiting were all significantly lower in the TEAS group (Table 3).

Table 2 Intra-operative consumption of anaesthetics and clinical recovery from anaesthesia in patients receiving transcutaneous electrical acupoint stimulation (TEAS) or sham treatment. Values are mean (SD).

	TEAS (n = 33)	Sham (n = 32)	p value
Remifentanyl; μg.kg ⁻¹ .min ⁻¹	0.06 (0.02)	0.09 (0.03)	0.01
Propofol; μg.kg ⁻¹ .min ⁻¹	109.1 (15.8)	108.9 (18.0)	0.95
Length of recovery room stay; min	35.6 (12.9)	48.3 (16.3)	0.01
LMA removal time; min	10.2 (2.5)	17.8 (4.4)	0.01
Reorientation; min	14.6 (3.2)	26.5 (5.0)	0.01

LMA, laryngeal mask airway.

During operation, we noticed that three patients showed decreased blood pressure (mean < 60 mmHg) and four exhibited decreased heart rate (< 50 beats.min⁻¹) in the sham group. Compared with patients in TEAS group, there was no significant difference. Otherwise, intra-operative haemodynamics were stable in both groups (Table 4).

As shown in Table 5, there were no significant differences between groups at any time point with regard to concentrations of adrenaline, noradrenaline, ACTH, blood glucose and cortisol.

Discussion

In this study, we have demonstrated that the length of stay in the recovery room, remifentanyl consumption, time to removal of the LMA and time to reorientation were all significantly shorter in the patients who were pre-treated with TEAS. Pain was less and other side-effects were also fewer.

The beneficial effect of electroacupuncture or TEAS has also been shown in earlier trials [8–11]. However, the parameters and acupoints used to administer the stimulation vary among different studies. According to the theory of traditional Chinese medicine, the LI4 acupoint, which belongs to the Large Intestine Meridian of Hand-yangming, is considered to exert an analgesic and sedative effect and is usually a key acupuncture point for anaesthesia in head and neck surgery [18]. The ST36 acupoint is one of the most important acupoints on the Foot Yang Ming stomach meridian; it can regulate the spleen and stomach and can reduce vomiting [19]. Stimulation of the PC6 acupoint, which belongs to the Hand-jueyin pericardium meridian, was reported to decrease myocardial ischaemia in patients with coronary heart disease [20] and also to alleviate postoperative nausea and vomiting after laparoscopic surgery [21]. Applying acupuncture to multiple points achieves greater effects than any single one. Therefore, in our current trial, the acupuncture points Hegu (LI4), Zusanli (ST36) and Neiguan (PC6) were selected. The dense-disperse frequency of 2/10 Hz was chosen based on our previous studies [9, 13]. Further, blood pressure and heart rate were stable in both groups, bispectral index values were controlled at between 40 and 55 for all patients, and the hormonal assays suggested similar surgical

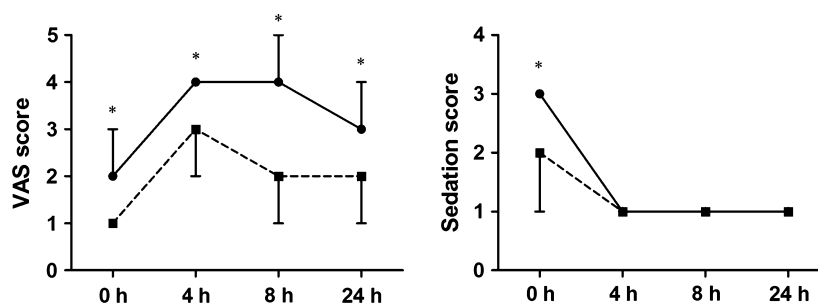


Figure 3 Median postoperative visual analogue (VAS) pain (left) and sedation scores (right) in patients receiving transcutaneous electrical acupoint stimulation (■) or sham treatment (●). Error bars are IQR. * $p < 0.05$.

Table 3 Intra- and postoperative side-effects in patients receiving transcutaneous electrical acupoint stimulation (TEAS) or sham treatment. Values are number (proportion).

	TEAS (n = 33)	Sham (n = 32)	p value
Intra-operative			
Hypotension	4 (12.1%)	3 (9.4%)	0.72
Bradycardia	4 (12.1%)	4 (12.5%)	0.96
Postoperative			
Dizziness	6 (18.2%)	17 (53.1%)	0.01
Nausea	7 (21.2%)	15 (46.9%)	0.03
Vomiting	4 (12.1%)	11 (34.4%)	0.03
Pruritus	2 (6.1%)	8 (25.0%)	0.03

stress levels in both groups. Patients were not informed of the allocation, and we used placebo gel electrodes in the same therapeutic position for patients in the sham group, to make them less distinguishable from those who actually received the stimulation. Thus, our study's strengths include adequate blinding, standardised TEAS and anaesthesia protocols, and comparable anaesthesia depth and surgical stress levels.

We recognise the limitations of our present trial. Due to the nature of the intervention, it was not possible to blind acupuncturists to treatment. Although the acupuncturist conducting TEAS were not masked to treatment assignment, none participated in data acquisition and analysis. In addition, the patients collected are from a single centre; large-scale multicentre clinical trials are still needed.

Ambulatory surgery guidelines strongly recommend earlier recovery from anaesthesia to ensure patient safety and improve the quality of recovery. Like other trials [22, 23], our data showed that although propofol consumption was similar in both groups, TEAS reduced the intra-operative consumption of remifentanyl by almost 35%, suggesting that TEAS may exert an analgesic effect. Accordingly, the times to removal of the LMA and reorientation were shorter in the TEAS group; that is to say, they regained consciousness and cognitive and communicative ability sooner. We suggest that this is because of the lower doses of remifentanyl that they received. Likewise, the

Table 4 Intra-operative heart rate (HR), mean arterial pressure (MAP), end-tidal carbon dioxide partial pressure (P_{ETCO_2}) and bispectral index (BIS) monitor values in patients receiving transcutaneous electrical acupoint stimulation (TEAS) or sham treatment. Values are mean (SD).

	TEAS (n = 33)				Sham (n = 32)			
	HR; beats.min ⁻¹	MAP; mmHg	P_{ETCO_2} ; kPa	BIS	HR; beats.min ⁻¹	MAP; mmHg	P_{ETCO_2} ; kPa	BIS
T ₀	75.5 (8.8)	85.5 (7.8)	—	96.7 (2.4)	76.8 (10.4)	85.4 (6.8)	—	97.4 (2.2)
T ₁	72.8 (8.6)	86.8 (6.9)	—	95.6 (2.9)	75.5 (9.7)	86.2 (7.6)	—	96.5 (2.1)
T ₂	64.3 (8.9)	83.4 (5.1)	4.7 (0.4)	54.6 (2.2)	67.5 (12.4)	89.5 (8.7)	4.7 (0.5)	56.5 (2.7)
T ₃	59.1 (13.2)	83.7 (7.9)	4.3 (0.6)	43.7 (10.8)	61.4 (12.1)	80.3 (9.3)	4.6 (0.5)	44.8 (12.1)
T ₄	62.9 (9.8)	77.5 (8.3)	4.3 (0.6)	53.7 (10.4)	63.1 (8.7)	78.5 (10.3)	4.4 (0.6)	54.0 (12.8)
T ₅	79.3 (8.9)	103.2 (10.5)	—	81.4 (9.7)	83 (11.6)	100.5 (12.3)	—	79.8 (8.5)

T₀, baseline; T₁, at the end of TEAS; T₂, at the beginning of surgery; T₃, 30 min after the start of surgery; T₄, at the end of surgery; T₅, 5 min after removal of laryngeal mask airway.

Table 5 Hormone levels in patients receiving transcutaneous electrical acupoint stimulation (TEAS) or sham treatment. Values are mean (SD).

	TEAS (n = 33)				Sham (n = 32)			
	T ₀	T ₂	T ₃	T ₄	T ₀	T ₂	T ₃	T ₄
Adrenaline; ng.ml ⁻¹	26.4 (4.9)	26.6 (5.2)	28.3 (4.7)	29.5 (6.5)	27.2 (5.4)	25.4 (5.7)	27.6 (4.5)	29.0 (6.6)
Noradrenaline; pg.ml ⁻¹	313.4 (72.1)	312.7 (89.5)	320.5 (94.4)	320.4 (87.3)	319.9 (87.9)	318.2 (78.7)	325.6 (85.4)	316.5 (80.3)
ACTH; pg.ml ⁻¹	32.2 (9.7)	30.6 (8.4)	31.3 (7.8)	33.5 (9.5)	28.4 (8.4)	31.4 (6.9)	30.6 (7.4)	31.8 (8.6)
Blood glucose; mmol.l ⁻¹	5.2 (0.8)	5.4 (0.6)	5.6 (0.7)	5.6 (0.5)	5.4 (0.6)	5.3 (0.5)	5.6 (0.6)	5.4 (0.7)
Cortisol; ng.ml ⁻¹	29.6 (6.6)	28.2 (7.4)	27.1 (7.6)	30.5 (8.7)	28.3 (5.6)	25.7 (7.0)	26.6 (6.5)	29.7 (8.1)

ACTH, adrenocorticotrophic hormone; T₀, baseline; T₂, at the beginning of surgery; T₃, 30 min after the start of surgery; T₄, at the end of surgery.

reduction in the side-effects of anaesthesia we observed confirms the results of previous trials [11, 12] and is also of course particularly welcome in ambulatory surgery. Transcutaneous electric acupoint stimulation also achieved better analgesic effects within the first 24 h postoperatively, adding weight to a previous study suggesting analgesic benefit from auricular acupuncture before hip arthroplasty [22]. It is well known that the analgesic mechanism of acupuncture is related to endogenous opioid system activation [24]. Previous studies have shown that electroacupuncture promoted enkephalin release in animal experiments and clinical trials [24, 25]. Collectively, the findings suggest that electroacupuncture alleviates pain by stimulating endogenous opioid release. However, several clinical trials found that electroacupuncture did not reduce pain or analgesia requirements after surgery, and had no effect on postoperative nausea [26, 27]. This may be because the electroacupuncture used in the studies was during the intra-operative period [27], which may not be appropriate for facilitating neurotransmitter release [28]. Further, the authors selected a dense-disperse high frequency of 2/100 Hz, or a single frequency of 10 Hz but a dense-disperse frequency of 2/10 Hz [26, 27].

Overall, the findings from this trial suggest that pre-operative TEAS is useful in shortening recovery room stay and also in improving the quality of patient recovery after general anaesthesia for outpatient surgery. Compared with acupuncture or electroacupuncture techniques, TEAS is a non-invasive technique and more 'user friendly', and can potentially be applied by any

anaesthesiologist or medical personnel with minimal training. Thus, as a simple and practical approach, TEAS-assisted anaesthesia should be used more widely in ambulatory surgery after further studies to identify the most beneficial combination of time, frequency and intensity of stimulation.

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Competing interest

No competing interests declared.

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